Ever-Present Noise

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Key Concepts

1) Noise and interference obscure the information in signals.

2) Noise generally refers to contamination that has little structure and is always present due to thermal and quantization effects.

3) Interference generally refers to contamination that has some structure and is often of man-made origin.

4) Definition and identification of what is signal, interference, and noise is problem specific.

5) The quality of a measurement is often described using ratios of signal power to noise or interference power, such as signal-to-noise ratio (SNR).

6) Signal processing methods exploit differences in the characteristics of the signal, interference, and noise to mitigate the effects of the noise and interference on the information in the signal.

Transcript

Ever-Present Noise

Introduction

Noise and interference obscure the information in signals. Extracting the information of interest in the presence of noise and interference is often the primary purpose of signal processing.

The distinction between noise and interference is blurry in general. The term “noise” often refers to contamination having a random or unpredictable nature, while the term “interference” is often used for contamination with structure or predictable aspects. Interference may even be very similar to the signal of interest.

For example, suppose you are driving an automobile and conversing with the person in the passenger seat. The conversation between people in the back seat or the sound from the radio would typically be considered interference. It has a lot of structure, and is similar to the sound spoken by your partner. Sounds due to wind turbulence around the car, the tires rolling on the pavement, and other cars nearby would be considered noise and have a more random character.

Noise

Noise is always present.

First of all, if a physical signal is sensed or conveyed by electrical means, then the random thermal motion of electrons in the electronic circuitry will introduce noise into the measurement. In very high performance applications the electronics may be cooled to reduce thermal noise.

Second, recall from our previous lesson that analog signals must be converted to digital form so they can be manipulated in a computer. This conversion requires representing the value of the signal using limited precision, that is, a limited set of discrete values.

The step of rounding the analog value of the signal to the nearest digital value introduces an error and is called quantization. Quantization error represents a noise as it causes the values of the signal to deviate from their true values.

Thermal and quantization noise are inherent in all applications that use electronics for measuring signals and digital computers for processing them.

Other types of noise may be present in the environment, such as the wind turbulence mentioned in the automobile example or cosmic microwave background radiation due to the big bang.

Interference

Interference is often of man-made origin. It may always be present in some applications, never be present in others, or be intermittent.

Interference may be inherent to the application. In wireless communication systems, such as cellular telephony, limited electromagnetic spectrum and high demand force many users to share the same frequency band and potentially interfere with each other.

It may be an artifact of the environment, such as when power line interference contaminates a measurement.

Interference may be deliberately introduced to "blind" an adversary in a military scenario. For example, one party may purposely transmit very high power microwave signals to overwhelm their adversary's radar system and interfere with its ability to detect attacking aircraft.

It's all relative

The definitions of signal, interference, and noise depend on your frame of reference.

In the automobile example of the introduction, the front seat passengers regard the conversation between back seat passengers as interference. However, a person in the back

seat regards the conversation between front seat passengers as interference.

If you are assessing fetal heart health, then the maternal electrocardiogram (heart signal) is interference. On the other hand, the maternal electrocardiogram is the signal of interest if you are addressing maternal heart health. Truly, one person's signal is another person's interference.

Ambiguity between signal and noise may be less common than that between signal and interference, but is still possible. For example, cosmic microwave background radiation is noise if you wish to receive communication from a satellite, but is signal if you are a cosmologist studying the evolution of the universe.

Quantifying noise and interference

The impact of noise and/or interference on the information contained in a signal depends in general on how big the signal is relative to the noise and/or interference. Hence, quantities like signal-to-noise ratio (SNR), signal-to-interference ratio (SIR) or signal-to-interference-and-noise ratio (SINR) are used to describe the quality of a signal.

SNR, SIR, and SINR are normally defined as the ratio of average signal power to average noise (or interference, or interference plus noise, respectively) power. This ratio is then usually expressed in decibels (dB).

If Ps is the signal power and Pn the noise power then the SNR in dB is 10*log10(Ps/ Pn). Note that if root mean square values of signal and noise (or interference) are used, then the conversion to dB is based on 20*log10 of the ratio of root mean square values.

Acceptable values of SNR may vary widely depending on the application. For example, in a wireless communication application an SNR of 5 dB may be more than adequate, while a high-fidelity audio application may require SNR well above 65 dB, a factor of more than a million greater.

Signal processing strategies

Usually the primary purpose of signal processing is to minimize the impact of noise and interference on the information conveyed by the signal. Specific approaches for doing so form the basis for your future studies of this field.

Signal processing methods generally exploit differences between the characteristics of the signal of interest and those of contaminating noise and interference. For example, if the signal of interest repeats and the noise is random in nature, averaging repeated measurements tends to attenuate the noise by averaging out fluctuations while reinforcing the signal.

Mathematical models or descriptions for signals, interference, and noise form the basis for systematic approaches to signal processing algorithm selection and

design. The role of mathematical modeling in signal processing is the topic of our next lesson.

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